



SKG14 IC8 Pacing & Emerging Technologies for ICM Engineers,L2

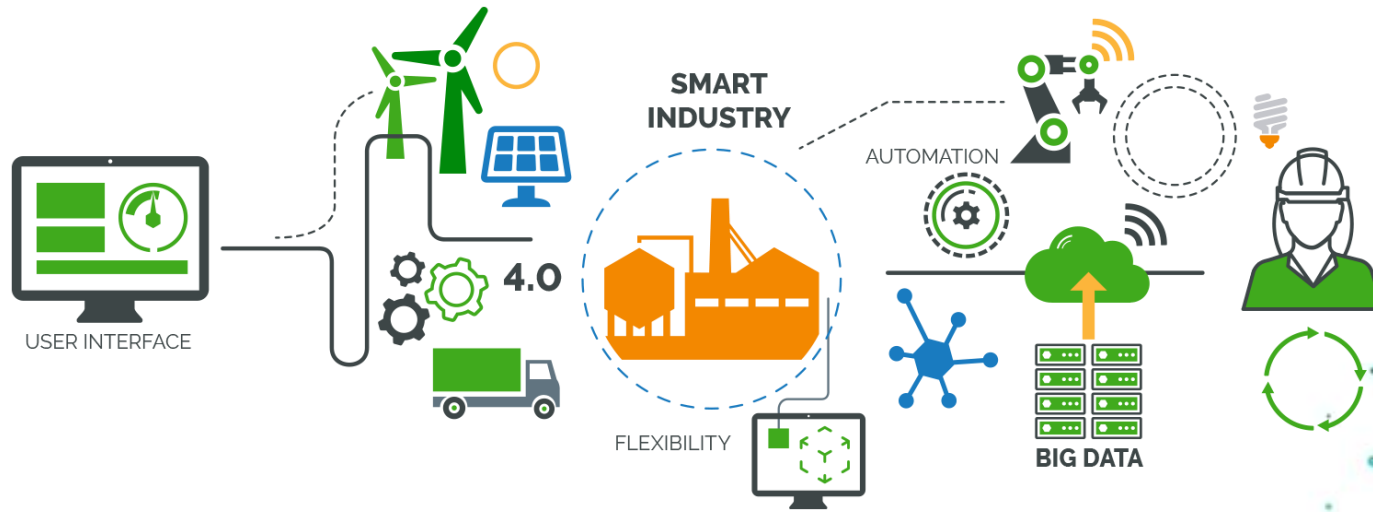
6.0 IIoT Hands-on Activities

July 2025

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6.1 Group Discussion

Instrument Efficiency Cases in Oil and Gas

Objectives

- To gauge the understanding of learners utilizing the building blocks of IOT to develop an architecture

Instructions

- Learners to split into 4 groups
- Each group will be given same 3 scenario covering
 - IIoT – Oil and Gas
 - IoT – Oil and Gas
 - IoT – Smart Home
- Group will be given 30mins to discuss
- Build a complete infrastructure that solves the business pain points utilizing earlier learning experience
- Develop the infrastructure in a single PowerPoint slide
- Each group will be given 15 mins to present to the audience

Note:

Do not concentrate too much on technology and make sure to include all the possible building block

Instrument Efficiency Cases in Oil and Gas

Business Pain points or challenges



Enable Early Intervention (avoid COVID-19 cluster via instantaneous contact tracing)



Project Continuity (even under MCO and threat of COVID-19)



Productivity Monitoring (movement tracking and maximize activities)



Cost Avoidance (productivity loss, project delay)



Safety Monitoring (instant location during emergency)

Instrument Efficiency Cases in Oil and Gas

Secure Personnel Online Tracking (SPOT)

SPOT Analytics

Dashboard and visualization to provide real-time personnel location, insights, and perform contact tracing. Capable of identifying all individuals who are in close proximity with the PUI over the last 2 weeks, within 5 minutes, using data collected from SPOT App and Tag.

Data Sent over via Wi-Fi and GPS (RSSI) to determine location and proximity

SPOT Tag and App

Low power device to detect workers location using GPS and Wi-Fi.

App installed on Android to detect workers location

Platform Tier

Network Tier

Edge Tier



Instrument Efficiency Cases in Oil and Gas

Scenario:

An oil production plant is looking to improve its monitoring and maintenance processes. They decide to deploy IIoT vibration sensors on critical equipment such as pumps and compressors and gauges monitoring.

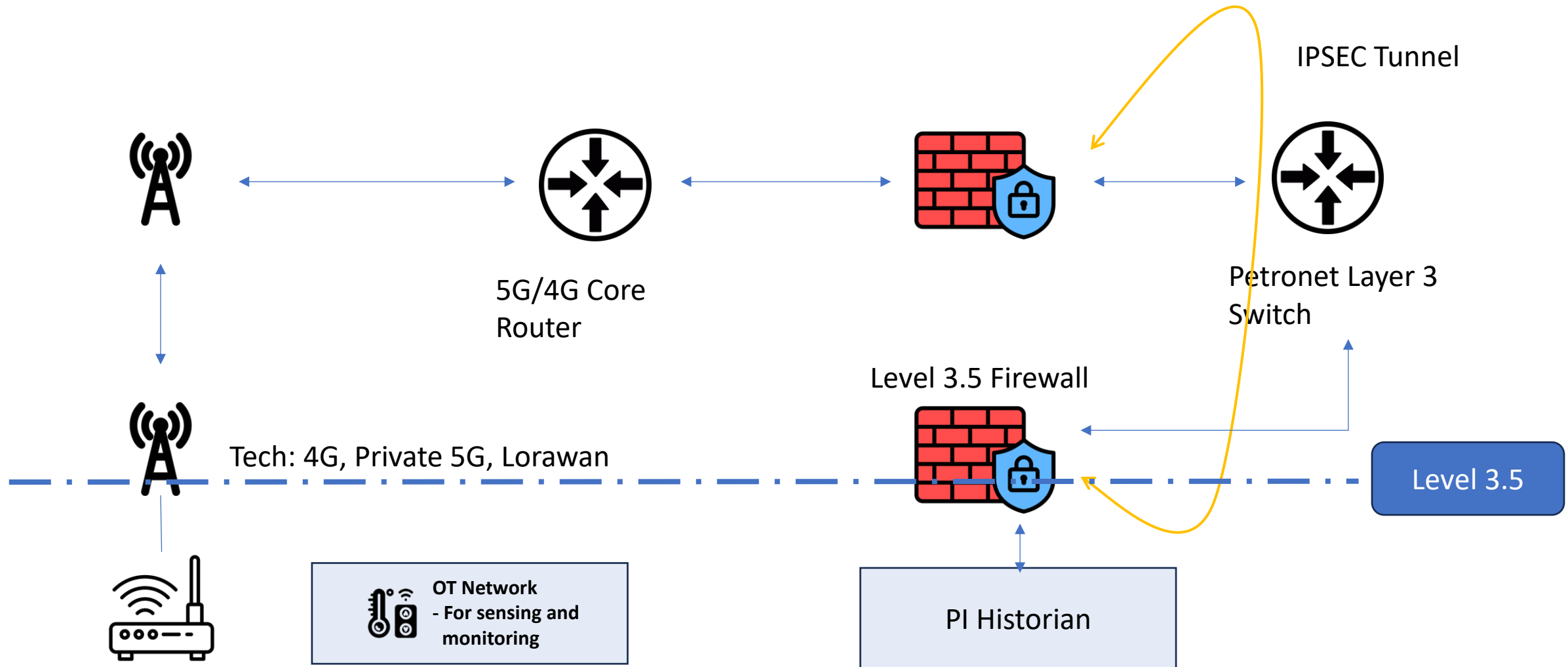
These sensors and gauges monitoring continuously monitor vibration levels and gauge meters and detect anomalies that could indicate potential equipment failures. The data collected by these sensors is then relayed to a PI historian system, which stores and analyzes the information in real-time.

The offshore oil production plant is located 1KM away from onshore plant.

Describe how do IIoT sensors collect and transmit data to the PI historian system?

What are some protocols and technologies are used to for end-to-end connectivity?

Instrument Efficiency Cases in Oil and Gas



IoT Smart Home Use Case

Scenario: Smart Home Automation

The family is planning to visit their hometown for Raya and will be away for a week. They want to ensure their home is secure and their four pet cats are well taken care of. Here's how their smart home setup can help:

Perimeter Security:

1.Smart Security Cameras: Install cameras around the perimeter of the house. These cameras can detect motion and send real-time alerts to the family's smartphones if any unusual activity is detected.

2.Smart Doorbell: A smart doorbell with a camera and intercom system allows the family to see and communicate with anyone at the door, even when they are away.

3.Window and Door Sensors: These sensors can detect if any windows or doors are opened and send alerts to the family.

Lighting:

1.Smart Lights: Set up smart lights to turn on automatically at sunset and turn off at sunrise. This can be done using a schedule or based on the local sunset and sunrise times.

2.Motion-Activated Lights: Install motion-activated lights around the exterior of the house to deter intruders.

Pet Care:

1.Automated Pet Feeder: Use a smart pet feeder that can dispense food at specified times. The feeder can be controlled remotely via a mobile app, allowing the family to adjust feeding times and portions as needed.

2.Smart Water Dispenser: Ensure the cats have access to fresh water with a smart water dispenser that refills automatically.

3.Pet Monitoring Camera: Install a camera in the area where the cats spend most of their time. This allows the family to check on their pets and ensure they are safe and comfortable.

IoT Smart Home Use Case

